DF^{2}RQ: Dynamic Feature Fusion via Region-Wise Queries for Semantic Segmentation of Multimodal Remote Sensing Data

Shiyang Feng, Zhaowei Li, Bo Zhang, and Bin Wang, *Senior Member, IEEE*

***Abstract*— Although remote sensing (RS) data with multiple modalities can be used to significantly improve the accuracy of semantic segmentation in RS data, how to effectively extract multimodal information through multimodal feature fusion remains a challenging task. Specifically, existing methods for multimodal feature fusion still face two major challenges: 1) due to the diverse imaging mechanisms of multimodal RS data, the boundaries of the same foreground may vary across different modalities, leading to the inclusion of unwanted background semantics in the fused foreground features, and 2) RS data from different modalities exhibit varying discriminative abilities for different foregrounds, making it challenging to determine the proportion of semantic information for each modality in the fusion results. To address the above issues, we propose a dynamic feature fusion method based on region-wise queries, namely, DF^{2}RQ, for SS of multimodal RS data. This method is primarily composed of two components: the spatial reconstruction (SR) module and the dynamic fusion (DF) module. Within the SR module, we propose an SR scheme that samples foreground features from different modalities, achieving independent reconstruction of different unimodal features, thereby alleviating the semantic mixing between foreground and background across modalities. In the DF module, a feature fusion scheme based on unimodal feature reference positions is proposed to obtain fusion weights for each modality, thereby enabling the DF of complementary features from multiple modalities. The performance of the proposed method has been extensively evaluated on various multimodal RS datasets for SS, and the experimental results consistently show that the proposed method achieves state-of-the-art (SOTA) accuracy on multiple commonly used metrics. In addition, our code is available at https://github.com/I3ab/DF2RQ.**

***Index Terms*— Feature fusion, multiple modalities, remote sensing (RS) data, semantic segmentation.**

I. Introduction

REMOTE sensing (RS) data comprise various modalities, including RGB, hyperspectral imaging (HSI), synthetic aperture radar (SAR), and digital elevation model (DEM). Precisely interpreting RS data for these modalities significantly benefits many applications, including urban planning [1], environmental monitoring [2], and disaster assessment [3]. In various interpretation tasks of RS data, semantic segmentation is a fundamental and important task, which refers to assigning semantic labels to each pixel in the RS data [4], [5], [6]. Over the past few years, significant efforts have been made in designing semantic segmentation methods for single-modal (unimodal) RS data [7], [8], [9]. However, unimodal RS data may not adequately express the complete semantics of the foregrounds in complex RS scenes, thereby limiting the accuracy of semantic segmentation tasks.

The advancements of RS imaging technology have made large-scale, spatially aligned multimodal RS data publicly available [10], [11], [12]. Furthermore, multimodal RS data generated by various imaging techniques can capture various attributes of RS scenes, including texture structure, spectral reflectance, and elevation details [13]. Specifically, RS data in the RGB modality are generated by the signals of the red, green, and blue bands by optical sensors. It can provide rich texture and color information but lacks detailed spectral information and may be affected by weather and lighting conditions [14], [15]. As for HSI, it is capable of capturing data with high spectral resolution across a wide electromagnetic spectrum range and can accurately identify foregrounds based on their unique spectral characteristics. However, its coarse spatial resolution limits its further applications [16], [17]. Concerning SAR, it is generated by irradiating the target surface with microwave pulses and then capturing the backscattered signals. SAR can penetrate cloud cover and acquire data under any lighting conditions, but it is susceptible to speckle noise and may suffer from geometric distortions due to the incidence angle of the radar waves on sloped surfaces [18], [19]. Regarding DEM, it quantifies the surface height of targets by using LiDAR or radar sensors, thereby providing topographic information by representing terrain elevations. However, DEM is susceptible to interference from dense vegetation and lacks semantic information about the target [20], [21]. Overall, RS data from different modalities are distinguished by their inherent benefits and limitations, and the characteristics of multimodal data of the same RS scene can be observed from the examples in Fig. 1, which can be briefly summarized as follows: 1) the boundaries of identical foregrounds might differ across modalities and 2) the discriminative capacity of RS data from different modalities

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Shiyang Feng, Zhaowei Li, and Bin Wang are with the Key Laboratory for Information Science of Electromagnetic Waves (MoE), Fudan University, Shanghai 200433, China (e-mail: wangbin@fudan.edu.cn).

Bo Zhang is with Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China.

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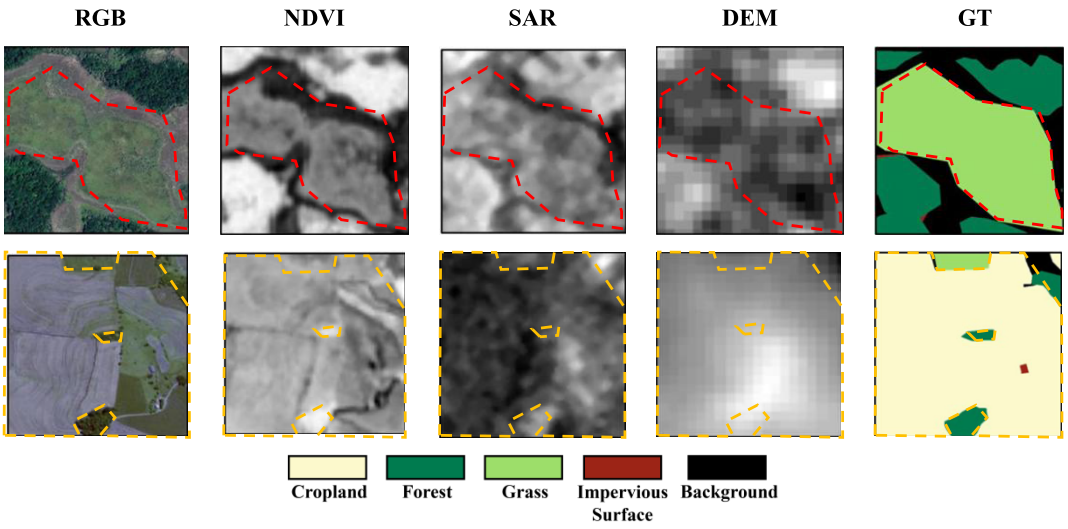


Fig. 1. Multimodal data of the same RS scene. The samples are from the Globe230k dataset [10], which includes four modalities: RGB, NDVI, SAR, and DEM. The examples in the first row indicate that the boundaries of the same foreground may differ across different modalities, while the examples in the second row demonstrate that the discriminative capabilities of RS data under various modalities may vary among different foregrounds.

may vary for different categories of foregrounds. Therefore, utilizing the complementary information between different modalities of RS data to improve the accuracy of semantic segmentation remains a challenging task.

Several methods have been proposed to integrate multimodal RS data for semantic segmentation. Among these methods, feature fusion methods have been widely adopted due to their robust multimodal representation capabilities and superior generalization capabilities, where multimodal features are derived through the fusion of unimodal features [22], [23], [24]. However, most of these methods are specifically designed for RS data with only two modalities, which are difficult to adapt to or expand to the needs of more modalities. Therefore, to accommodate situations involving two or more modalities, summation and concatenation are two common strategies in feature fusion methods.

For the methods based on summation [25], [26], the unimodal features are summed and normalized to obtain the fused feature. Such methods provide a simple and straightforward strategy for feature fusion, but all modalities are assumed to be of equal importance, which may result in weak representation ability of the fused features or cause modality competition [27], [28]. In terms of the methods based on concatenation [11], [29], [30], [31], the unimodal features are first concatenated along the channel dimension and then mapped through a few linear or convolutional layers to reduce the channel dimension. Compared to the summation-based methods, the concatenation-based methods preserve the distinctive information of each modality and enhance the learning of interactions between modalities. However, the concatenation-based methods are incapable of dynamically adjusting the proportion of information from different modalities according to the current samples, and the fused features obtained through these methods may also suffer from the risk of overfitting [32], [33], which hinders the complete utilization of complementary information across different modalities and thus limiting the accuracy of downstream tasks.

To address the above issues, we propose a dynamic feature fusion method based on region-wise queries for semantic segmentation of multimodal RS data. Specifically, the proposed method is mainly composed of two components: a spatial reconstruction (SR) module and a dynamic fusion (DF) module. In the SR module, the unimodal features extracted by different backbone networks are reconstructed separately. Since the boundaries of the same foreground may be spatially inconsistent across different modalities, we propose a deformable reconstruction scheme for unimodal features to mitigate the mixing of semantics from foreground and background. In the DF module, the unimodal features reconstructed by the SR module are represented as a series of reference points in a polar coordinate system. To obtain fused features from the unimodal features, the contribution of each unimodal feature to the fused feature is quantified by measuring the distance between the reference points of the unimodal features and the predicted point representing the fused feature. In this manner, the fusion weights for features of different modalities can be dynamically adjusted based on the current sample, ensuring that the complementary information between modalities is maximally incorporated into the fused feature, even when the number of modalities is arbitrary, such as more than two or even more.

Extensive experiments are conducted on various multimodal RS data for semantic segmentation tasks. All experimental results consistently demonstrate that the proposed method can obtain discriminative robustness representations of multimodal RS data, thereby significantly improving the accuracy of multimodal RS semantic segmentation tasks. Furthermore, based on a simple but strong baseline model, a series of ablation studies are conducted to validate the effectiveness and generalization of the SR module and the DF module in our method.

The main contributions of this article can be briefly summarized as follows.

1) We propose a feature fusion method (named DF^{2}RQ) for the semantic segmentation of multimodal RS data.

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By dynamically fusing unimodal features from different modalities, the proposed method achieves state-of-the-art (SOTA) results on various multimodal RS datasets for semantic segmentation. It is also the first proposed method for multimodal feature fusion applicable to RS datasets containing two or more modalities.

2) In the SR module, we propose an SR scheme for unimodal features, which prevents the fusion of foreground and background features caused by the misalignment of foreground boundaries in different modalities, thus improving the representation quality of the fused features.
3) In the DF module, we propose a region-wise feature fusion scheme based on the distance relationship between unimodal features and fused features, which enables the dynamic integration of complementary information from various modalities and is versatile enough to be applied to scenarios with two or more modalities.

The remainder of this article is organized as follows. A comprehensive overview of the related works is given in Section II. In Section III, the proposed method is presented in detail. Section IV presents the experimental results and conducts an insightful analysis of our proposed methods. Concluding remarks are presented in Section V.

II. Related Works

A. Semantic Segmentation for Unimodal Data

As a critical task in computer vision, semantic segmentation has made significant progress in recent years. The methods for semantic segmentation of unimodal data are applicable to a variety of contexts, including natural scenes, RS scenes, and more.

Semantic segmentation methods in natural scenes [5], [6], [34], [35], [36], [37], [38] can be categorized into CNN-based and transformer-based methods. For the methods based on CNNs, FCN [34] is proposed to implement semantic segmentation in an end-to-end paradigm. To address the substantial impact of changes in foreground scale on semantic segmentation performance, UPerNet [39] is proposed to aggregate feature maps of different scales. Furthermore, the dilated convolution operator is utilized in Deeplab [5] to expand the receptive field, enhancing the capture of long-range spatial relationships. Nevertheless, these methods still do not fully capture the contextual information, which is pivotal for enhancing semantic segmentation accuracy. For the methods based on transformers [40], self-attention and cross-attention mechanisms have been widely used in the encoder and decoder structures of semantic segmentation methods [6], [41], thus further improving the accuracy and generalization ability of semantic segmentation methods in natural scenes.

Compared to the natural scenes, data of RS scenes depict more complex scenarios characterized by a larger number of small objects and smaller semantic distinctions between foreground and background. To alleviate the imbalance between the foreground and background as well as the issue of false alarm, a foreground-aware framework is proposed in FarSeg [42] to capture the foreground semantics from relation- and optimization-based perspectives. Besides, a disentangled learning-based segmentation method [43] is proposed to address the significant intraclass variance issues. Furthermore, an edge-weighted affinity loss is proposed in NPALoss [44] to improve the segmentation accuracy of small objects and foreground boundaries. With the rise of large-scale pretrained models, Mamba-based [45] and SAM-based [46] methods have been proposed and have achieved impressive performance for segmentation tasks in RS data. A boundary-enhanced dual-stream network is proposed in BEDSN [47] to address the issues of intraclass inconsistency and boundary blur in RS images. This network demonstrates promising performance in both quantitative and visual assessments while maintaining low model complexity. Although semantic segmentation methods with unimodal data have achieved remarkable results, unimodal data are insufficient to completely capture the semantic information of the foreground, which limits the performance of semantic segmentation methods in various complex RS scenarios.

B. Semantic Segmentation for Multimodal Data

Semantic segmentation methods for multimodal data incorporate three key stages: unimodal feature extraction, multimodal feature fusion, and semantic segmentation decoding. Among these stages, multimodal feature fusion plays a crucial role in transforming unimodal features into integrated multimodal representations and has been extensively studied in the context of both natural and RS scenes.

There are various methods suitable for multimodal feature fusion in natural scenes. For RGB and depth (RGB-D) data, gating mechanisms and residual unit are used in GRBNet [48] to compute the complementarity of different unimodal features, and then, the unimodal features are aggregated into multimodal features. To integrate features from LiDAR data and RGB images from different perspectives in autonomous driving scenarios, several serial cross-attention layers are adopted to fuse feature tokens from different modalities in BEVFormer [49] and DeepInteraction [50]. As for most vision-language methods [51], [52], [53], [54], [55], [56], [57], unimodal features are first concatenated along the spatial dimension and then processed through multiple transformer blocks to obtain multimodal representations.

Unlike the natural scenes, data from different modalities are usually spatially aligned in the RS scenes. Therefore, feature fusion methods based on summation and concatenation are used in vFuseNet [26] and HighDAN [11]. To further enhance the multimodal feature representation, a gate fusion module is proposed in CMGFNet [58], which determines the weights for feature fusion based on the impact of different modalities on the segmentation results. Furthermore, a cross-modal multiscale transformer and a multiscale context augmented transformer are used in ABCNet [59] and CMFNet [60] to fuse multiscale feature maps from different modalities. Besides, the intuition-inspired hypergraph network [61] is proposed to model high-order relationships in multimodal RS data. Moreover, the multimodal multiscale feature fusion method was first proposed in FTransUnet [62], achieving impressive accuracy in multimodal RS image segmentation tasks.

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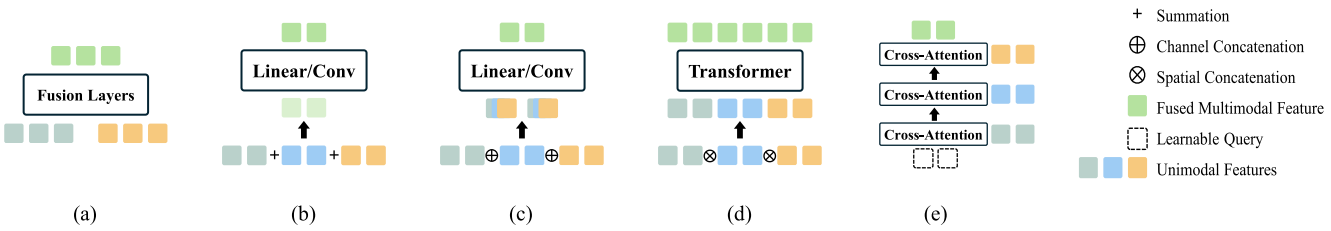


Fig. 2. Schematic of the differences in multimodal feature fusion methods. (a) Feature fusion methods for two modalities. (b) Summation-based methods [25], [26]. (c) Concatenation-based methods [11], [29], [30], [31]. (d) Parallel attention-based methods [50], [51], [52]. (e) Serial attention-based method [48], [49].

Overall, as a crucial stage in the semantic segmentation of multimodal data, most of the previous feature fusion methods are designed for multimodal data with only two modalities, and the exploration of fusion methods applicable to any number of modalities remains limited and challenging. The differences among various multimodal feature fusion methods can be illustrated in Fig. 2.

III. Proposed Method

Generally, the semantic segmentation method for multimodal RS data mainly consists of three stages: unimodal feature extraction, multimodal feature fusion, and semantic segmentation decoding. Among these stages, multimodal feature fusion is crucial as it integrates complementary features from diverse modalities, thereby enhancing the performance of the segmentation tasks. However, most of the previous methods focus on the fusion method of features with two modalities, and the strategies for integrating features from more modalities are mainly based on summation and concatenation. Consequently, we propose a dynamic feature fusion method based on region-wise queries for semantic segmentation of multimodal RS data, namely, DF^{2}RQ. The overall structure of the proposed method is shown in Fig. 3.

First, the unimodal features of multimodal RS data are extracted by the unimodal backbone networks. Subsequently, the unimodal features are fused by the proposed DF^{2}RQ. Specifically, the proposed DF^{2}RQ mainly consists of two modules, which are the SR module and the DF module. In the SR module, the single-modal features from different modalities are regionally reconstructed based on a group of spatial biases. In the DF module, the reconstructed unimodal features are fused based on the distance relationships between reference points within a polar coordinate system to obtain multimodal features. Finally, the multimodal features are decoded by the semantic segmentation decoding head to obtain the segmentation results.

A. Unimodal Feature Extraction

RS data from various modalities often exhibit variation in the number of channels. For example, RGB images consist of three channels, hyperspectral images may include several hundred channels, and DEM data generally contains only a single channel. However, even though existing feature extraction networks [60], [62] can effectively extract and integrate information from RS data of different modalities and scales, they typically lack the capability to simultaneously process inputs with a varying number of channels and cannot benefit from the backbone networks pretrained on large-scale datasets.

In order to obtain the unimodal features of multimodal RS data, the feature extraction network φ^m of each modality is composed of two parts: 1) stem blocks and 2) backbone networks F_b^m. Specifically, the stem block F_s^m consists of two 1×1 convolution layers, which are used to map the input from different modalities into embeddings with the same number of channels. Then, the backbone networks F_b^m with unshared weights are used to obtain the unimodal features of different modalities. Denote the samples of each modality as $x^m \in \mathbb{R}^{H \times W \times d^m}$; then, the multimodal RS data are $X = \{x^m | m \in \{A, B, C, \ldots\}\}$, and the modality number of X is $M = |m|$. The process of obtaining unimodal features from multimodal data can be briefly represented as follows:

$$f^m = F_b^m\left(F_s^m\left(x^m\right)\right) = \varphi^m\left(x^m\right) \tag{1}$$

where $F_s^m(\cdot)$ is used to extract the embeddings e^m of x^m and F_b^m is used to map e^m to the unimodal features $f^m \in \mathbb{R}^{H' \times W' \times D}$.

To obtain the multimodal features of multimodal RS data, the unimodal features are fused through the multimodal feature fusion stage described in Sections III-B–III-D.

B. Region-Wise Query Generation

The proposed DF^{2}RQ is a multimodal feature fusion method based on query vectors. As a key part of the overall feature fusion module, the generation strategy of query vectors is described as follows.

Since the distribution of foregrounds in different categories is random in the spatial regions of RS data and RS data in different modalities have distinct discriminative capacities for the foregrounds of various categories, we adopt a strategy of generating query vectors in a region-wise manner, that is, the region-wise query vectors are only used for SR and DF within their corresponding region. The detailed process of region-wise query generation is shown in Fig. 4.

Define the region-wise query vectors as $Q = \{q_0, q_1, \ldots, q_i, \ldots | q_i \in \mathbb{R}^{M \times D}\}$, and the number of query vectors is N_q. Then, the region size S_{q_i} of each query q_i is determined by N_q and the spatial size $S_f = (H', W')$ of the unimodal features f^m

$$S_{q_i} = \left(\frac{H'}{N_q}, \frac{W'}{N_q}\right). \tag{2}$$

To obtain the region-wise query vectors from the unimodal features f^m, the unimodal features are concatenated along the

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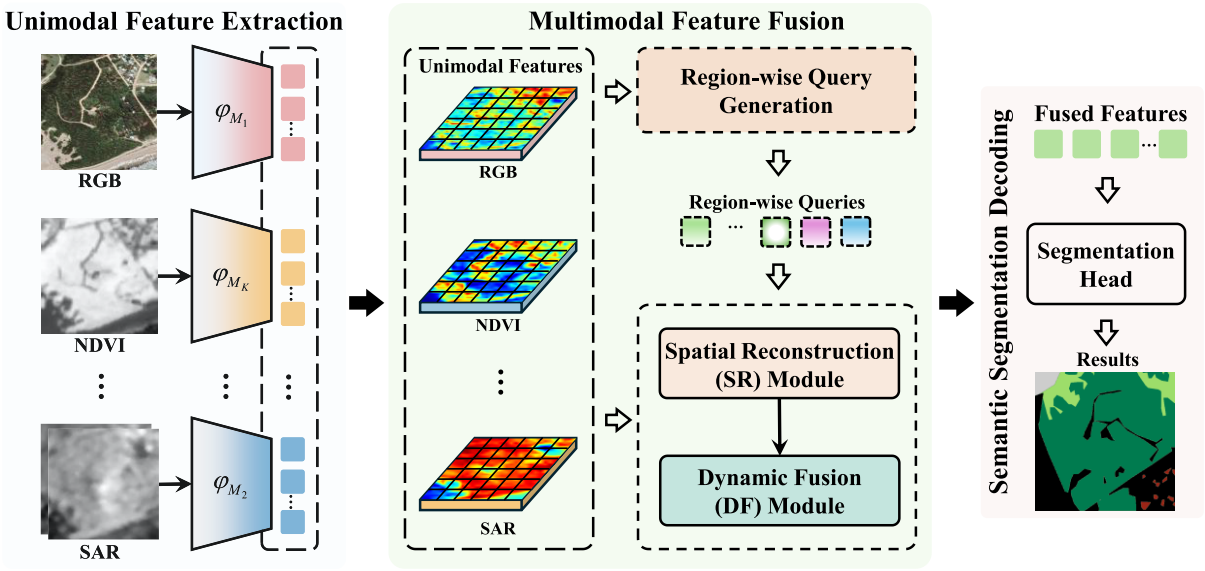


Fig. 3. Detailed structure of the proposed method. The framework of semantic segmentation method for multimodal RS data mainly consists of three stages: unimodal feature extraction, multimodal feature fusion, and semantic segmentation decoding. Here, we propose a feature fusion method named DF²RQ specifically for the multimodal feature fusion stage.

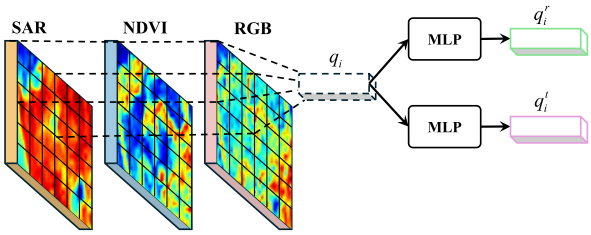


Fig. 4. Process of the region-wise query generation. Here, RGB, SAR, and NDVI data from Globe230k [10] are taken as an example to illustrate the overall process.

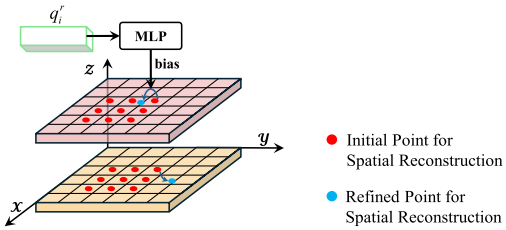


Fig. 5. Structure of the SR module. The SR module is used to reconstruct unimodal feature to avoid the fusion of foreground and background semantics.

channel dimension first, and then, the query vectors for each region are extracted by an average pooling layer. Thus, the query vector q_i for the spatial region i is derived as follows:

$$q_i = \text{AvgPooling}\left(f_i^A \oplus f_i^B \oplus f_i^C \oplus \cdots\right) \tag{3}$$

where $\oplus$ denotes channel concatenation. Subsequently, the query vector q_i of each region i is mapped to $q_i^r \in \mathbb{R}^D$ and $q_i^t \in \mathbb{R}^D$ by two different MLPs

$$q_i^r = \text{MLP}^r(q_i) \tag{4}$$

$$q_i^t = \text{MLP}^t(q_i) \tag{5}$$

where both MLP^r and MLP^t consist of two linear layers with ReLU activation, and the weights of MLP^r and MLP^t are not shared.

C. SR Module

Considering the spatially misaligned boundaries of foreground features in different modalities, the proposed SR module is used to reconstruct the unimodal features, so as to avoid the mixing of foreground and background semantics in the subsequent feature fusion modules. The structure of the SR module is shown in Fig. 5.

Inspired by the feature reconstruction methods [63], [64], [65] based on the deformable receptive field, we propose a query-based feature reconstruction strategy in the SR module that does not require direct supervision information. Specifically, for the feature map f^m of each modality, the feature map in a region i on f^m is $f_i^m \in \mathbb{R}^{(H'/N_q)\times(W'/N_q)\times D}$, and the initial positions of the feature vectors in f_i^m can be represented as a set of initial points

$$P_i^m = \{(\alpha_0, \beta_0), (\alpha_1, \beta_1), \ldots\}. \tag{6}$$

In order to perform the SR for each feature vector in f_i^m, the query vector $q_i^r \in \mathbb{R}^D$ is mapped through an MLP to the spatial bias ΔP_i^m for each initial point

$$\Delta P_i^m = \text{MLP}\left(q_i^r\right) = \{(\Delta\alpha_0, \Delta\beta_0), (\Delta\alpha_1, \Delta\beta_1), \ldots\} \tag{7}$$

where the MLP is composed of two linear layers, each using ReLU and tanh as activation functions.

Then, the refined points are obtained by adding the coordinates of the initial points to the corresponding spatial biases, and then, the reconstructed features are sampled from the positions of these refined points. Since the 2-D coordinates of the refined points are usually floating-point numbers, bilinear interpolation is employed to calculate the features reconstructed from these refined points.

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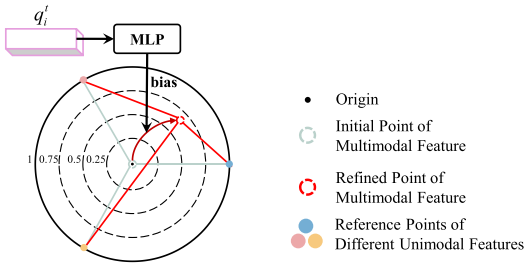


Fig. 6. Relationship between unimodal and multimodal features in polar coordinates. Here, the number of modalities is set to three.

According to the above scheme, the reconstructed unimodal features f'^m are obtained region by region from f^m. Besides, it is worth noting that the location of a refined point and its corresponding original point may not be in the same region, thus making the receptive field of the reconstructed features more flexible.

D. DF Module

Like the reconstruction scheme of the unimodal features f^m in the SR module, the DF process of the multimodal features $f^{\text{fused}} \in \mathbb{R}^{H' \times W' \times D}$ is also conducted in a region-wise manner in the DF module. To obtain discriminative and informative representations of multimodal RS data with two or more modalities, the relationships between f^{fused} and f^m need to be described by an effective and general strategy. Moreover, the discriminative ability of f^m varies for different foregrounds, and the relationships between f^{fused} and f^m should be dynamic.

Due to the above reasons, the feature fusion module needs to possess two fundamental characteristics: 1) features from different modalities are equally important before fusion and 2) the importance of each modality's features should dynamically change based on the input data. To design such a feature fusion strategy in the DF module, the relationships between the fused multimodal features f^{fused} and unimodal features f^m are described through a series of reference points, and the relationships between f^{fused} and f^m within the DF module are illustrated in Fig. 6.

To simplify the explanation, the fusion process of unimodal features in a region i will be described next, and the same process will be applied to the other regions. Furthermore, the coordinates of the reference points can be expressed in different forms. However, due to the inconvenience in intuitively determining the positions of reference points for different modalities in Cartesian coordinates, the polar coordinate form is adopted to uniformly represent the coordinates of reference points across different modalities. In addition, the unimodal features used for fusion in the DF module are $f_i'^m$ that reconstructed by the SR module.

Specifically, the reconstructed unimodal features $f_i'^m$ in region i are represented as a series of reference points$U = \{u_i^m | m \in \{A, B, C, \ldots\}\}$, where the reference point of $f_i'^m$ is $u_i^m = (\rho^m, \theta^m)$. For a different region i, the reference point u_i^m of $f_i'^m$ is the same. Besides, the coordinates of the reference points U in the polar coordinate system are uniformly preset to fixed values. For example, when the number of modalities is three, the coordinate of the reference points is

$$u^A = (\rho^A, \theta^A) = (1, 0) \tag{8}$$

$$u^B = (\rho^B, \theta^B) = \left(1, \frac{2}{3}\pi\right) \tag{9}$$

$$u^C = (\rho^C, \theta^C) = \left(1, \frac{4}{3}\pi\right). \tag{10}$$

In general, the coordinates of the reference points for an arbitrary number of modalities are defined as

$$\rho^m = 1 \tag{11}$$

$$\theta^m = \frac{2\pi}{M} j, \quad j \in \{0, \ldots, M-1\}. \tag{12}$$

In order to obtain the proportion of each unimodal feature $f_i'^m$ in the multimodal features f_i^{fused}, the coordinate of the point that represents f_i^{fused} is defined as u_i^{fused}, and the initial coordinate of u_i^{fused} is set to the origin u_o

$$u_o = (\rho_o, \theta_o) = (0, 0). \tag{13}$$

The above settings about the reference coordinates of unimodal features and the initial coordinates of multimodal features ensure that the unimodal features have the same importance by default. Then, the relationships between f_i^{fused} and $f_i'^m$ can be dynamically expressed by the distance relationship between u_i^{fused} and u_i^m. Concretely, the polar bias $\Delta u_i^{\text{fused}}$ is predicted from the query vectors q_i^t by an MLP

$$\Delta u_i^{\text{fused}} = \left(\Delta\rho_i^{\text{fused}}, \Delta\theta_i^{\text{fused}}\right) = \text{MLP}\left(q_i^t\right) \tag{14}$$

where the MLP is composed of two linear layers, each using ReLU and tanh as activation functions.

Then, u_i^{fused} is obtained as follows:

$$u_i^{\text{fused}} = \left(\rho_i^{\text{fused}}, \theta_i^{\text{fused}}\right) \tag{15}$$

$$\rho_i^{\text{fused}} = \Delta\rho_i^{\text{fused}} + \rho_o, \quad \theta_i^{\text{fused}} = \Delta\theta_i^{\text{fused}} + \theta_o. \tag{16}$$

The Euclidean distances ω_i^m between u_i^{fused} and u_i^m can be calculated as follows:

$$\begin{aligned}\omega_i^m &= \left\| u_i^{\text{fused}} - u_i^m \right\| \\ &= \sqrt{\left(\rho_i^{\text{fused}}\right)^2 + \left(\rho_i^m\right)^2 - 2\rho_i^{\text{fused}}\rho_i^m \cos\left(\theta_i^{\text{fused}} - \theta_i^m\right)}.\end{aligned} \tag{17}$$

The distance ω_i^m is normalized to $[0, 1]$ by the SoftMax function and then used as the weights for the fusion of unimodal features in region i. The fused multimodal features of region i are obtained as follows:

$$f_i^{\text{fused}} = \sum_m \omega_i^m f_i'^m. \tag{18}$$

Finally, the multimodal features f^{fused} are obtained region by region according to the above strategy.

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E. Decoding Head and Loss Function

1) Semantic Segmentation Decoding: The decoding of multimodal features f^{fused} is jointly accomplished by the segmentation head and the auxiliary head. For the segmentation head, UperNet [39] is adopted to decode multimodal features f^{fused} of different scales, enabling accurate segmentation of foregrounds at different scales in multimodal RS data. For the auxiliary head, FCN [34] is employed to assist in achieving better accuracy and more stable convergence during the model training process.

2) Loss Function: The loss function L for model training includes the segmentation loss L_{seg} corresponding to the segmentation head and the auxiliary loss L_{aux} corresponding to the auxiliary head. Specifically, cross entropy is used as L_{seg}, and FocalLoss [34] is used as L_{aux}. Overall, the loss function L can be obtained as follows:

$$L = L_{\text{seg}} + \lambda L_{\text{aux}} \tag{19}$$

where λ is the balance coefficient of auxiliary loss.

IV. Experimental Results and Analyses

A. Dataset Description

Four public datasets are used to evaluate the performance of the proposed method, including Globe230k [10], C2Seg-BW [11], Vaihingen [66], and Potsdam [66].

1) Globe230k: The Globe230k dataset [10] consists of four modalities: RGB, NDVI, SAR (VV, VH), and DEM with three, one, two, and one bands, respectively. It comprises 232 819 annotated images categorized into ten foreground classes: "Cropland," "Forest," "Grass," "Shrub," "Wetland," "Water," "Tundra," "Impervious Surface," "Bareland," and "Ice/snow." Moreover, the "Background" category was not included in the accuracy evaluation and segmentation visualization according to the official setup in Globe230k [10]. Besides, the ratio of training set, validation set, and test set is 7:1:2. Since the spatial resolution varies across different modalities, the patch size of data from different modalities are all uniformly resized to 512 × 512. Furthermore, the annotated images originate from various locations around the world, covering an area of over 60 000 km^2, demonstrating significant variability and diversity.

2) C2Seg-BW: The C2Seg-BW dataset [11] is a part of cross-city multimodal datasets [11], providing multimodal RS data for the cities of Beijing and Wuhan in China, and labeled with 13 foreground categories, namely, "Surface Water" (SW), "Street Network" (SN), "Urban Fabric" (UF), "Industrial, Commercial, and Transport" (ICT), "Artificial Vegetated Areas" (AVA), "Arable Land" (AL), "Permanent Crops" (PC), "Pastures" (P), "Forests" (F), "Shrub" (S), "Open Spaces with no vegetation" (OS), and "Inland Wetlands" (IW). Moreover, the dataset includes hyperspectral, multispectral, and SAR data, with 116, 4, and 2 bands, respectively. Following the benchmark in HighDAN [11], the "Background" (BG) category was not included in the accuracy evaluation and segmentation visualization. Moreover, the patch size of the multimodal RS data is set to 256 × 256, and the numbers of samples in the training set and test set are 7140 and 850, respectively. In addition, C2Seg-AB (Augsburg and Berlin) is also included in the C2Seg dataset, but due to the small size of this subset (273 training samples and 140 test samples), we did not conduct an experiment and analysis on it.

3) Vaihingen: The Vaihingen dataset [66] is a widely used benchmark for semantic segmentation of high-resolution RS data provided by the ISPRS and released in 2014. The dataset contains NearInfRared-Red-Green (NIRRG) and DSM data of Vaihingen and manually annotated ground truth data for five different foreground classes: "Impervious Surface," "Building," "Low Vegetation," "Tree," and "Car." This dataset comprises a total of 33 true orthophoto data (TOPs) with a spatial resolution of 0.09 m per pixel and width dimensions ranging from 1887 to 3816 pixels. To evaluate the performance of the proposed method, it is divided into a training set with 16 images and a testing set with 17 images. Besides, the original images are cropped into nonoverlapping 512 × 512 patches for the training and testing process.

4) Potsdam: The Potsdam dataset [66] is another used benchmark for semantic segmentation of high-resolution RS images, released in 2018. The dataset contains infrared–red–green–blue (IRRGB) and DSM data of Potsdam, and it has the same manually annotated classes as the Vaihingen dataset. The data cover an area of 38.4 km^2 at a spatial resolution of 0.05 m. Compared with the Vaihingen dataset, the Potsdam dataset has a larger scale with 38 TOPs data and a larger spatial size with 6000 × 6000 pixels. Following the common practice, the dataset is divided into a training set of 24 images and a test set of 14 images. Like the Vaihingen dataset, each original data in the Potsdam dataset is also cropped into nonoverlapping 512 × 512 patches.

B. Implementation Details

1) Baseline Model: To validate the effectiveness of our proposed method, we build a simple but strong baseline model. As with the proposed method, the baseline model also consists of three stages: unimodal feature extraction, multimodal feature fusion, and semantic segmentation decoding. The feature extraction networks of unimodal are consistent with Section IV-A and the ViT-Base [41] that pretrained on ImageNet1k is adopted as the backbone networks. This allows for better representation of unimodal features from different modalities compared to training from scratch. Moreover, when the number of modalities changes, only new branches of the feature extraction network need to be added without extensive modifications to the network structure. For the multimodal feature fusion, the unimodal features f^m are first concatenated along the channel dimensions and then mapped to f^{fused} by an MLP that consists of two linear layers with ReLU activation. The semantic segmentation head and the corresponding loss functions follow Section IV-E. It is worth noting that the baseline model can be applied to scenarios involving both multimodal and unimodal scenarios, with the baseline model in unimodal scenarios not including a multimodal feature fusion stage.

2) Experimental Settings: We implement our model based on the Pytorch and mmsegmentation framework. Since the official mmsegmentation framework cannot flexibly support

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training multimodal data with two or more modalities, we developed a series of basic multimodal data augmentation methods for training all models, including random resize, random crop, and random flip. Without explicitly specifying, all the models are trained 80 000 iterations on eight NVIDIA A100 GPUs (80G) with a batch size set to 32. The balance coefficient λ of auxiliary loss is set to 0.4. The optimizer is AdamW with the initial learning rate set to 0.00006, and the learning rate adjusted using a cosine annealing.

3) Evaluation Metrics: To evaluate the segmentation results of the multimodal RS data, the overall accuracy (OA), the mean $F1$ score (m$F1$), and the mean intersection over union (mIoU) are employed. These standard statistical indices can provide fair comparisons of the performance of the proposed method and other SOTA methods. Furthermore, the $F1$ score and IoU are first calculated for each foreground class, and then, the average values (m$F1$ and mIoU) of these metrics across different foreground classes are obtained.

C. Experimental Results

We conducted a comparative evaluation of the proposed DF^{2}RQ with current SOTA methods on multiple datasets of multimodal RS data. Due to the varying number of modalities across different datasets and the fact that most previous methods are designed for multimodal data with two or three modalities, there are significant differences in the SOTA methods across various datasets. The SOTA methods involved in comparison across different datasets are as follows.

For the latest and largest Globe230k [10] dataset, several representative semantic segmentation methods (i.e., DeepLab [5], Segformer [6], FCN [34], OCNet [35], PSPNet [36], PointRend [37], SwinTransformer [38], and ViT [41]) in natural scenes are adopted to compare with our baseline model in RGB modality. Moreover, the proposed baseline model is also used to compare with our DF^{2}RQ.

For the C2Seg-BW [11] dataset, FastFCN [67] and Segformer [6] are representative methods in the field of semantic segmentation, based on CNN and transformer, respectively, both demonstrating high accuracy in natural image scenes. Moreover, domain shift between samples in different cities is considered in the HighDAN [11], AdaptSeg [68], DSAN [69], and DualHR [70] to further improve the accuracy of semantic segmentation task.

For the Vaihingen and Potsdam datasets [66] with only two modalities, vFuseNet [26] and CMGFNet [58] consider multiscale information in the feature fusion process to improve the segmentation accuracy, and ABCNet [59], CMFNet [60], and FtransUnet [62] effectively fuse different modal features based on attention mechanism. Besides, the comparison on the Vaihingen dataset also includes the latest and advanced unimodal methods (i.e., the Mamba-based [45] and SAM-assisted [46] methods, and BEDSN [47]).

Moreover, few studies have conducted experiments on the latest Globe230k [10] and C2Seg-BW [11] datasets, and the experimental results of the SOTA methods for these datasets are derived from their respective literature [10], [11]. As for the classical Vaihingen and Potsdam datasets [66], the experimental results of the SOTA methods are all derived from the respective papers of each method.

1) Quantitative Results:

a) Results on Globe230k Dataset: We present the experimental results on the Globe230k [10] dataset for fair comparison with the SOTA methods in Table I. We first compare the baseline model with a series of classical semantic segmentation algorithms on RGB modality. The experimental results show that our unimodal baseline model provides powerful semantic segmentation accuracy despite its simplicity. Moreover, we use the same baseline model to conduct unimodal training on NDVI, SAR, and DEM modalities. The experimental results indicate that the unimodal baseline model trained with RGB achieved the highest accuracy, followed by the results obtained from SAR and NDVI, which were quite similar, while the model trained with DEM achieved the lowest accuracy. Therefore, in our experiments with unimodal baseline models, we present the experimental results under both three-modal (i.e., RGB, SAR, and NDVI) and four-modal scenarios (i.e., RGB, SAR, NDVI, and DEM).

The experimental results indicate that the multimodal feature fusion method based on concatenation in the baseline model not only fails to improve the accuracy of semantic segmentation but also leads to the reduction of accuracy. Especially when the DEM modality is introduced, the accuracy of the multimodal baseline model further decreases. This phenomenon can be attributed to the RGB modality providing sufficient semantic information for most foregrounds in Globe230k, where introducing semantic information from other modalities into the fused features using a concatenation-based method may lead to a decrease in downstream task accuracy. It is also notable that this phenomenon is consistent with the observation in other multimodal learning tasks [28], [33].

In order to validate the effectiveness of the proposed multimodal feature fusion method, we further compare the proposed method with the multimodal baseline model. The experimental results show that the accuracy of the proposed method is not only significantly better than the multimodal baseline model but also surpasses the best unimodal baseline model. Moreover, even when the DEM modality with lower resolution and less semantic information is included, the segmentation accuracy is not negatively affected. Besides, the proposed method achieves good segmentation accuracy on most foreground categories, which further demonstrates the superiority of the proposed method. Besides, there are significant differences in accuracy among various methods when it comes to the "Tundra" category. This is likely attributable to the small sample size within this category [10].

b) Results on C2Seg-BW Dataset: The experimental results on the C2Seg-BW dataset are shown in Table II. The mIoU, OA, and m$F1$ of the proposed method are 15.13%, 46.22%, and 21.19%, respectively, which are significantly higher than those of the previous methods. Compared to semantic segmentation methods (i.e., Segformer [6] and FastFCN [67]) that do not consider cross-domain gaps between different cities, the proposed method demonstrates improved

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TABLE I

EXPERIMENTAL RESULTS ON THE GLOBE230K DATASET. THE ACCURACY OF EACH CATEGORY, MIOU, OA, AND m$F1$ METRICS ARE GIVEN IN THE TABLE. THE VALUE IN BOLD DENOTES THE BEST RESULTS IN EACH COLUMN. THE EXPERIMENTAL RESULTS OF THE SOTA METHODS ARE OBTAINED FROM THE LATEST GLOBE230K DATASET [10]

Method	Modality	Category Accuracy										mIoU	OA	mF1
		Cropland	Forest	Grass	Shrub	Wetland	Water	Tundra	Impervious Surface	Bareland	Ice/snow			
FCN [34] (2015)	RGB	90.73	95.37	71.90	70.20	68.23	88.34	0.30	94.03	59.59	96.26	67.97	89.12	/
OCNet [35] (2018)	RGB	89.98	89.71	75.34	57.58	49.62	82.67	27.91	80.11	74.78	91.45	70.12	87.45	/
DeepLabv3+ [5] (2018)	RGB	90.89	95.10	77.46	75.74	70.35	95.33	0.00	**94.29**	90.77	93.43	70.56	90.46	/
PSPNet [36] (2017)	RGB	91.41	94.91	77.32	74.32	77.62	93.93	34.21	90.47	90.35	97.51	73.12	90.49	/
PointRend [37] (2020)	RGB	92.10	95.78	64.56	72.49	71.16	89.07	4.22	86.67	87.49	97.72	67.08	88.64	/
ViT [41] (2020)	RGB	78.83	89.12	55.84	53.81	47.63	82.87	0.00	80.59	73.19	91.19	65.31	87.70	/
Segformer [6] (2021)	RGB	92.65	94.64	55.14	56.13	41.96	92.95	0.00	92.36	90.05	95.27	62.64	86.69	/
SwinTransformer [38] (2021)	RGB	89.42	96.52	**79.33**	71.73	77.28	94.99	42.64	93.06	90.74	98.06	75.72	90.90	/
Baseline	RGB	92.27	97.02	76.32	79.49	80.62	94.40	29.89	92.32	94.28	**98.32**	77.28	91.93	85.67
	NDVI	87.63	89.98	31.00	25.35	29.77	89.13	0.00	85.46	77.41	75.11	49.56	77.83	68.07
	SAR	82.32	88.57	47.72	34.70	45.88	85.61	0.00	83.05	75.74	51.85	48.94	77.33	68.51
	DEM	69.26	78.14	2.12	1.07	4.06	76.26	0.00	1.34	48.72	40.36	23.57	57.33	36.18
Baseline (Concat)	RGB+NDVI+SAR	91.73	95.98	74.36	75.72	78.34	92.32	3.27	91.21	92.61	96.52	72.28	88.92	83.07
	RGB+NDVI+SAR+DEM	90.90	96.02	73.48	75.01	77.65	92.59	1.01	91.25	93.50	96.15	70.43	87.98	82.24
DF²RQ (Ours)	RGB+NDVI+SAR	94.54	97.00	75.54	**80.41**	81.01	95.57	68.75	92.90	95.00	97.49	**80.73**	92.63	88.69
	RGB+NDVI+SAR+DEM	**95.12**	**97.18**	75.37	79.91	**81.97**	**96.20**	**69.67**	93.92	**95.29**	98.26	80.41	**93.32**	**88.92**

TABLE II

EXPERIMENTAL RESULTS ON THE C2SEG-BW DATASET. THE ACCURACY OF EACH CATEGORY, MIOU, OA, AND m$F1$ METRICS IS GIVEN IN THE TABLE. THE VALUE IN BOLD DENOTES THE BEST RESULTS IN EACH COLUMN

Method	Modality	Category Accuracy													mIoU	OA	mF1
		SW	SN	UF	ICT	MDCS	AVA	AL	PC	P	F	S	OS	IW			
AdaptSeg [67] (2018)	HSI+MSI+SAR	59.57	**17.05**	25.06	32.77	1.94	8.99	9.11	0.30	0.00	32.98	0.00	0.00	0.00	8.92	29.26	14.44
DSAN [68] (2021)	HSI+MSI+SAR	0.00	0.00	**42.08**	25.68	1.22	9.55	**25.37**	0.00	0.00	46.85	0.00	0.00	0.00	7.09	18.55	11.60
DualHR [62] (2018)	HSI+MSI+SAR	60.51	0.29	0.76	24.19	0.26	4.19	0.00	0.00	0.00	32.70	0.22	0.33	0.01	6.17	31.97	9.53
SegFormer [6] (2021)	HSI+MSI+SAR	78.49	0.05	30.90	20.38	2.52	10.79	18.01	1.77	0.00	38.72	10.48	0.01	0.01	10.89	33.56	16.32
FastFCN [66] (2022)	HSI+MSI+SAR	45.39	2.38	38.44	27.63	0.86	8.83	0.00	0.00	0.00	4.14	0.00	0.00	0.00	5.96	21.22	9.82
HighDAN [11] (2023)	HSI+MSI+SAR	78.37	0.58	40.04	**43.67**	1.67	9.28	0.43	0.10	0.00	**47.22**	0.26	0.00	0.00	11.92	39.58	17.69
Baseline (Concat)	HSI+MSI+SAR	79.70	1.29	31.21	22.03	3.87	10.59	19.23	**2.62**	0.06	39.92	**12.02**	0.28	0.32	13.23	41.45	19.39
DF²RQ (Ours)	HSI+MSI+SAR	**80.14**	2.58	31.62	22.14	**5.42**	**12.26**	20.67	2.55	**1.21**	40.61	11.79	**1.67**	**1.88**	**15.13**	**46.22**	**21.19**

accuracy across all categories. Besides, it can recognize some difficult foreground categories (i.e., "P," "OS," and "IW"). In addition, due to significant semantic domain differences in certain categories (i.e., "SN," "UF," and "ICT") across different cities, the proposed method did not achieve the highest segmentation accuracy when compared with methods that consider domain differences, such as HighDAN and AdaptSeg.

c) Results on Vaihingen Dataset: As presented in Table III, the proposed DF²RQ shows significant improvements in terms of OA, m$F1$, and mIoU compared to the previous methods, indicating that the proposed DF²RQ successfully fused the complementary information from multiple modalities. Compared with existing SOTA methods, DF²RQ performs better in three categories, namely, "Tree," "Low Vegetation," and "Impervious Surface."

It is noteworthy that FtransUnet [62] achieves the highest accuracy among all compared methods in the "Building" category. This should be attributed to regions classified as "Buildings" typically exhibiting higher depth values, and FtransUnet is more focused on extracting multiscale features and leveraging depth information from DSM images. In contrast, our proposed method initially assumes equal importance for all modalities and predicts the importance of each modality based on the unimodal features of the current sample. Besides, to ensure a fair comparison with unimodal methods, this article directly adopts the experimental results from the latest SOTA methods (i.e., Mamba-based [45] and SAM-assisted [46] methods, and BEDSN [47]) as presented in their original papers. According to the experimental results, our method still demonstrates advantages in most overall metrics.

d) Results on Potsdam Dataset: Experiments on the Potsdam dataset for the proposed method also demonstrated results like those derived from the Vaihingen dataset. As presented in Table IV, the accuracy for "Car" and "Impervious Surface" of the proposed method is 98.24% and 95.02%, respectively, which amounts to a substantial increase of 1.55% and 1.11% compared to our multimodal baseline model. The mIoU, OA, and m$F1$ score values of our proposed

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TABLE III

EXPERIMENTAL RESULTS ON THE VAIHINGEN DATASET. THE ACCURACY OF EACH CATEGORY, MIOU, OA, AND M$F1$ METRICS IS GIVEN IN THE TABLE. THE VALUE IN BOLD DENOTES THE BEST RESULTS IN EACH COLUMN. THE CATEGORY ACCURACY OF BEDSN, RS^{3}MAMBA, AND FTUNETFORMER + SAM IS BASED ON THE IOU METRIC FROM THEIR RESPECTIVE PAPER [45], [46], [47]

Method		Modality	Category Accuracy					mIoU	OA	mF1
			Building	Tree	Low Vegetation	Car	Impervious Surface			
BEDSN [47] (2024)		NIRRG	89.95	80.56	71.99	75.92	85.71	80.82	90.14	89.25
RS^{3}Mamba [45] (2024)		NIRRG	93.83	83.66	67.84	81.97	86.62	82.78	/	90.34
FTUNetformer+SAM [46] (2024)		NIRRG	93.69	83.77	67.86	82.53	86.54	84.01	/	91.08
vFuseNet [26] (2018)		NIRRG+DSM	95.92	91.36	77.64	76.06	91.85	78.92	90.49	87.89
ABCNet [58] (2021)		NIRRG+DSM	95.20	89.70	84.50	85.30	92.70	81.30	90.70	89.50
CMGFNet [57] (2022)		NIRRG+DSM	97.75	91.60	80.03	87.28	92.35	82.26	91.72	90.00
CMFNet [59] (2022)		NIRRG+DSM	97.17	90.82	80.37	85.47	92.36	81.44	91.40	89.48
FTransUnet [61] (2024)		NIRRG+DSM	**98.20**	91.94	81.49	**91.27**	93.01	84.23	92.40	91.21
Ours	Baseline (Concat)	NIRRG+DSM	94.97	91.58	85.19	87.23	93.68	84.37	92.13	91.15
Ours	**DF^{2}RQ**	NIRRG+DSM	95.96	**93.61**	**87.06**	88.35	**93.97**	**85.02**	**92.87**	**91.33**

TABLE IV

EXPERIMENTAL RESULTS ON THE POTSDAM DATASET. THE ACCURACY OF EACH CATEGORY, MIOU, OA, AND M$F1$ METRICS ARE GIVEN IN THE TABLE. THE VALUE IN BOLD DENOTES THE BEST RESULTS IN EACH COLUMN. THE CATEGORY ACCURACY OF BEDSN [47] IS BASED ON THE IOU METRIC FROM ITS PAPER

Method	Modality	Category Accuracy					mIoU	OA	mF1
		Building	Tree	Low vegetation	Car	Impervious surface			
BEDSN [47] (2024)	IRRGB	91.64	75.66	75.20	90.53	84.49	75.27	89.17	84.08
vFuseNet [26] (2018)	IRRGB+DSM	97.23	84.29	89.03	95.49	91.62	84.26	90.22	91.26
ABCNet [58] (2021)	IRRGB+DSM	96.90	**89.10**	87.90	95.80	93.50	86.50	91.30	**92.70**
CMGFNet [57] (2022)	IRRGB+DSM	97.41	86.80	86.68	95.68	92.60	84.53	90.21	91.40
CMFNet [59] (2022)	IRRGB+DSM	97.63	87.40	88.00	95.68	92.84	85.63	91.16	92.10
FtransUnet [61] (2024)	IRRGB+DSM	97.78	88.27	88.48	96.31	93.17	86.20	92.40	92.41
Baseline (Concat)	IRRGB+DSM	**98.64**	88.57	**90.10**	96.69	93.91	87.35	92.23	92.66
DF^{2}RQ (Ours)	IRRGB+DSM	98.03	88.04	90.05	**98.24**	**95.02**	**88.12**	**92.48**	92.62

method are 88.12%, 92.48%, and 92.62%, respectively. Except for m$F1$, the other metrics of semantic segmentation task are significantly higher than the previous methods and the baseline model. Moreover, compared to the latest unimodal method (i.e., BEDSN [47]), multimodal methods generally achieve performance improvements. This enhancement should be attributed to more discriminative features in the semantic segmentation task, which originate from the fusion of different modality information.

2) Visualization Results:

a) Globe230k: Fig. 7 shows the visual results of semantic segmentation for ten semantic categories in the Globe230k dataset using two SOTA methods (i.e., Segformer [6] and SwinTransformer [38]) and the proposed method. According to the visual results, the proposed method achieves more accurate segmentation results in most foreground categories. Specifically, the visualization results in the first row of the figure demonstrate that the proposed method can accurately distinguish categories with high semantic similarity, such as "Grass" and "Cropland." The visualization in the second row shows that the method can accurately segment the edges of foreground categories, such as the "Water" category, while other methods exhibit more blurred foreground boundaries.

b) C2Seg-BW Dataset: Fig. 8 illustrates the visual results of semantic segmentation for ten semantic categories on the C2Seg-BW dataset using three SOTA methods (i.e., FastFCN [67], Segformer [6], and HighDAN [11]) and the proposed method. The first row of visualizations indicates that the proposed method achieves superior segmentation results for the "SW" and "SN" foreground categories; the second row demonstrates that the proposed method not only excels in the "SW" and "F" categories but also identifies locations of some "IW" category foregrounds.

The visualization results presented above demonstrate the superiority of our method in segmenting semantically similar categories and foreground objects, fully illustrating the effectiveness of the method.

Overall, the experimental results in Tables I–IV, along with the visualizations in Figs. 7 and 8, consistently indicate that our proposed method achieves competitive results on multimodal RS semantic segmentation datasets containing two or more modalities. However, limited by the number of modalities in the most recently available datasets, we have only conducted experiments with up to four modalities, and research on a wider variety of modalities still needs to be strengthened in future work.

In addition, as observed from Table I, if a single modality is sufficient to represent the semantic information of most foregrounds, merging other modalities by concatenation may actually degrade model performance. Besides, as shown in Tables II–IV, when significant complementarity exists between different modalities within the dataset, multimodal feature

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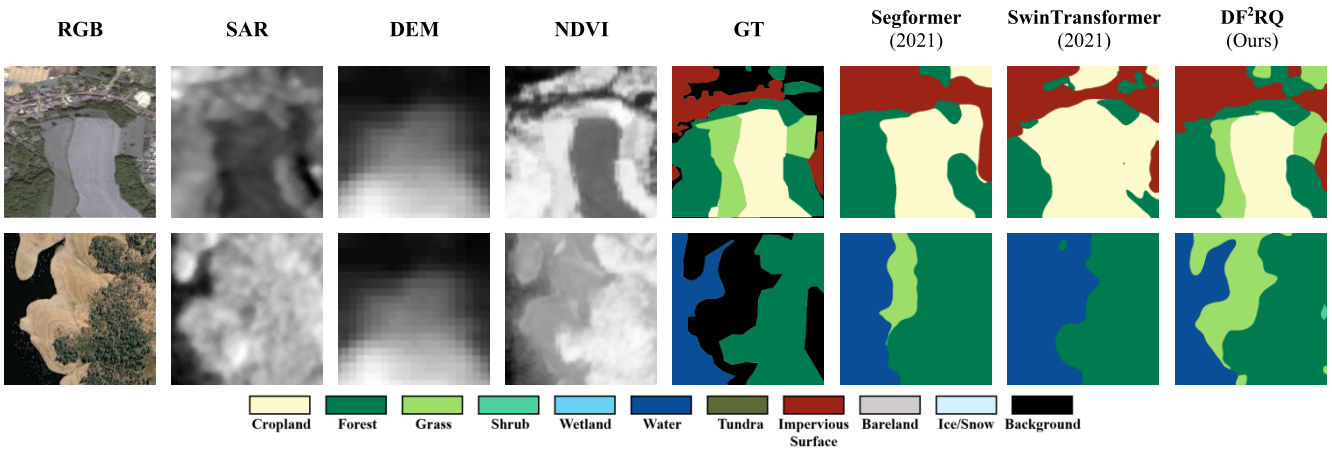


Fig. 7. Visualization of semantic segmentation results obtained from the Globe230k dataset by the proposed method.

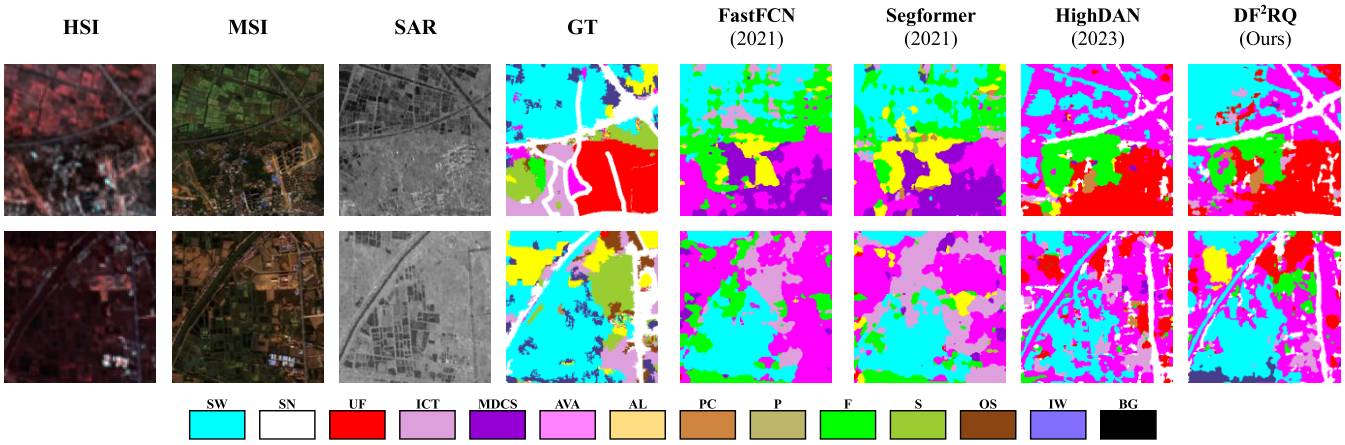


Fig. 8. Visualization of semantic segmentation results obtained from the C2Seg-BW dataset (Wuhan) by the proposed method.

fusion methods can enhance the accuracy of downstream tasks, even if the fusion is based on simple concatenation. Moreover, the baseline model outperforms most methods in terms of accuracy, which could be attributed to the more effective unimodal features extracted by the pretrained ViT-based backbone network.

D. Ablation Study

a) Effectiveness of SR and DF Modules: In the proposed method, the multimodal feature fuser is mainly composed of the SR module and the DF module. Thus, we conducted a series of ablation experiments on the Globe230k dataset to inspect the role of each module, and the experimental results are shown in Table V. For the SR module, it can be seen that the reconstruction of unimodal features can alleviate the mixing of semantic features from foreground and background, thus enhancing the discriminative ability of multimodal features and improving the accuracy of semantic segmentation task. For the DF module, the relationships between multimodal features and unimodal features are dynamically described, thereby ensuring that multimodal features can encapsulate the complementary information inherent in various unimodal features. Besides, even without employing the proposed DF module for feature fusion, the SR module can still enhance the performance of the concatenation-based fusion method in the baseline model, further illustrating the necessity of using the SR module to address the inconsistency in foreground scope across different modalities. Overall, the combination of the SR module and the DF module has achieved promising accuracy in the semantic segmentation task of multimodal RS data, which could be attributed to the regional and dynamic feature fusion strategy.

TABLE V
ABLATION STUDY ON THE EFFECTIVENESS OF THE SR MODULE AND THE DF MODULE

Method		mIoU	OA
SR	DF		
-	-	72.28	88.92
-	√	78.43	91.21
√	-	76.39	90.93
√	√	**80.73**	**92.63**

b) Comparison on Different Region Sizes of Region-Wise Query: The spatial size of each region is closely related to the receptive field of each query vector, which affects the granularity of unimodal feature reconstruction and multimodal feature fusion in the SR module and the DF module. Moreover, the number of query vectors will be affected by different

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TABLE VI
ABLATION STUDY ON DIFFERENT REGION SIZES OF REGION-WISE QUERIES

Spatial Size	mIoU	OA
1x1	79.73	91.88
2x2	**80.97**	**93.21**
4x4	80.73	92.63
8x8	79.21	92.11

TABLE VII
ABLATION STUDY ON DIFFERENT QUERY SOURCES

Query Source	mIoU	OA
RGB	79.92	92.21
NDVI	78.43	91.23
SAR	78.92	91.03
RGB+NDVI+SAR	**80.73**	**92.63**

region sizes when the spatial size of the feature map is fixed. Therefore, we conducted experiments under different region size settings, and the experimental results in Table VI show that different region sizes have a great impact on the segmentation accuracy. When the region size is too large, the fusion ratio of foreground features of different categories may remain consistent, resulting in insufficient representation ability of multimodal features of some categories. Conversely, when the region size is too small, the foreground in the same category may have different fusion weights, resulting in poor generalization ability of multimodal features. Overall, both above cases may lead to the degradation of segmentation accuracy. Therefore, we believe that the observed phenomena can be attributed to the variability of the distribution of different foreground classes in RS scenarios, as well as the difference in the proportion of modality information required to classify these categories.

c) Comparison on the Source of Region-Wise Query: The source of the query vectors determines the semantic information contained in the query vector, which affects the quality of feature reconstruction in the SR module and feature fusion in the DF module. Therefore, we compare the segmentation accuracy of our methods under different query sources. The experimental results in Table VII show that for unimodal features, RGB images contain the most abundant semantic information, and high segmentation accuracy can be achieved by only using RGB image features to generate query vectors. Furthermore, the segmentation accuracy is slightly improved by integrating the features from all modalities to generate query vectors. When the query vectors are generated only from modalities with less semantic information, the segmentation accuracy will be slightly decreased.

d) Model Size and Computational Complexity: We evaluate model size and computational complexity on the Vaihingen dataset using model parameter count and FLOPs metrics, respectively. The spatial size of the input data for testing FLOPs metrics is set to 256×256. As can be seen from Table VIII, the model size and computational complexity of multimodal models are significantly higher than those of unimodal models, which can be primarily attributed to the need for independent encoders for different modal data. Despite our model achieving the best results in terms of accuracy compared to other methods, it does not offer advantages in terms of model size or computational complexity. Besides, the advanced unimodal method (BEDSN [47]) achieves competitive accuracy with a very small model size, while CMGFNet [58] exhibits the lowest computational complexity along with favorable accuracy. Besides, compared to the multimodal baseline model, the proposed method achieves higher accuracy with fewer parameters and lower computational complexity, which should be attributed to the proposed region-wise queries-based feature fusion mechanism that enhances model efficiency.

TABLE VIII
ABLATION STUDY ON MODEL SIZE AND COMPUTATIONAL COMPLEXITY. FLOPS WERE EVALUATED UNDER THE CONDITION THAT THE SPATIAL SIZE OF INPUT DATA IS 256×256

Method	Modality	Accuracy		Efficiency	
		mIoU	mF1	Params(M)	FLOPs (G)
BEDSN [47] (2024)	NIRRG	80.82	89.25	**21.27**	83.89
RS3Mamba [45] (2024)	NIRRG	82.78	90.34	43.32	28.25
SAM-assisted [46] (2024)	NIRRG	84.01	91.08	96.14	50.84
CMGFNet [57] (2021)	NIRRG+DSM	82.26	90.00	64.2	**19.51**
CMFNet [59] (2022)	NIRRG+DSM	81.44	89.48	123.63	78.25
FTransUnet [61] (2024)	NIRRG+DSM	84.23	91.21	160.88	45.21
Baseline (Concate)	NIRRG+DSM	84.37	91.15	223.46	172.92
DF2RQ (Ours)	NIRRG+DSM	**85.02**	**91.33**	193.22	129.27

E. Insightful Analyses

The proposed method is mainly composed of the SR module and the DF module. By regionally reconstructing unimodal features in the SR module and dynamically fusing these features in the DF module, the resulting multimodal features have demonstrated strong discrimination and generalizability on multiple multimodal RS data, regardless of the number of modalities. Consequently, the proposed DF^{2}RQ has achieved significant improvements under various commonly used metrics on the semantic segmentation task of multimodal RS data. Based on the above experimental results, we provide the following insightful analyses.

1) *Effectiveness:* In the proposed DF^{2}RQ method, the SR module addresses the issue of semantic mixing between foreground and background across different modality features, while the DF module dynamically fuses features based on semantic relationships among modalities to serve downstream tasks. The experimental results also clearly demonstrate that, as a feature fusion approach for multimodal RS data, the DF^{2}RQ method exhibits superior accuracy in semantic segmentation tasks across various modalities of RS datasets.
2) *Scalability:* Previous multimodal feature fusion methods may face numerous challenges when dealing with data with more than two modalities [25], [26], [27], [28], [32], [33]. The proposed DF^{2}RQ, as a general and scalable approach, not only theoretically allows for easy adaptation to handle RS data with more modalities but has also been experimentally validated for its effectiveness in processing up to four modalities of RS data. Moreover, although the proposed method in this article is

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specifically applied to the semantic segmentation task of multimodal RS data and has been validated on multiple datasets, it can be easily adapted to various downstream tasks such as change detection and object detection by replacing the decoding head.

3) *Query Strategy:* As a multimodal feature fusion method based on region-wise query, we consider that the significance of query strategy is reflected in the following two aspects. First, variations in regional size result in features being divided into regions of varying quantities and sizes, thereby affecting the accuracy of feature reconstruction and fusion, and thus, the region size should be related to the foreground scale. Second, the initial source of the query vectors determines the richness of semantic information within the query vectors, thereby influencing the predictions of spatial biases and reference point biases in the SR module and DF module.
4) *Limitations:* Despite the advantages of the proposed method, it has three main limitations: 1) the fusion method is not applicable to RS scenarios involving text modality [57], despite the fact that text modalities are essential for enhancing the semantic information of RS data; 2) the method cannot integrate multimodal features of varying scales, potentially affecting the segmentation accuracy of foreground objects with smaller spatial dimensions; and 3) the feature extraction networks are independently designed for different modalities, leading to an increased model size and posing challenges for deployment.

V. Conclusion

This article has proposed a multimodal feature fusion method for semantic segmentation of multimodal RS data with any number of modalities, namely, DF²RQ. As a flexible and dynamic method, DF²RQ overcomes the limitations of previous methods designed specifically for multimodal RS data. To address the issue of semantic confusion between background and foreground caused by inconsistent boundaries of the same foreground in different modalities, the proposed method performs region reconstruction on unimodal features in the SR module to ensure the discriminative capability of multimodal features. Moreover, to dynamically integrate unimodal features from different modalities, the relationship between multimodal and unimodal features is represented as distances between different reference points in a polar coordinate system. The proposed method ensures dynamic modeling of the relationship between multimodal and unimodal features under any number of modalities. Consequently, the proposed DF²RQ achieves significant accuracy improvements in multimodal RS data segmentation, with its superior performance effectively validated across various public datasets.

Despite the promising results achieved by the proposed method, it still faces limitations such as being inapplicable to textual modalities, inability to integrate multiscale features, and high model size and complexity. Therefore, future research may focus on developing a unified framework that can integrate text modalities and also consider the fusion of multiscale features in different modalities to further improve the accuracy of downstream tasks, as well as the redesign of feature extraction networks for different modalities to reduce model complexity and ensure deployment feasibility.

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Shiyang Feng received the B.S. degree in automation from Donghua University, Shanghai, China, in 2020. He is currently pursuing the Ph.D. degree with the School of Information Science and Technology, Fudan University, Shanghai.

His research interests include object detection, semantic segmentation, and multimodal learning.



Zhaowei Li received the B.S. degree in electronic engineering from Fudan University, Shanghai, China, in 2022. He is currently pursuing the master's degree with the School of Information Science and Technology, Fudan University.

His research interests include semantic segmentation and multimodal remote sensing data fusion.



Bo Zhang received the Ph.D. degree in electronic science and technology from Fudan University, Shanghai, China, in 2022.

He is currently a Research Scientist with Shanghai AI Laboratory, Shanghai. He has published over 15 papers in top-tier international journals and conferences, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON MULTIMEDIA, *International Journal of Computer Vision,* CVPR, NeurIPS, and ACM-MM. His research interests primarily include multimodal structural representation learning, 2-D/3-D pretraining, and 2-D/3-D continuous learning.



Bin Wang (Senior Member, IEEE) received the B.S. degree in electronic engineering and the M.S. degree in communication and electronic systems from Xidian University, Xi'an, China, in 1985 and 1988, respectively, and the Ph.D. degree in system science from Kobe University, Kobe, Japan, in 1999.

After his graduation in 1988, he was with Xidian University as a Teacher. From 1999 to 2000, he was with the Communications Research Laboratory, Ministry of Posts and Telecommunications, Kobe, as a Research Fellow, working on magnetoencephalography signal processing and its application for brain science. As a Senior Supervisor, he was with the Department of Etching, Tokyo Electron AT Ltd., Tokyo, Japan, from 2000 to 2002, dealing with the development of advanced process control systems for etching semiconductor equipment. Since September 2002, he has been with the Department of Electronic Engineering, Fudan University, Shanghai, China, where he is currently a Full Professor and the Leader of the Image and Intelligence Laboratory. He has published more than 150 scientific articles in important domestic and international periodicals. He is the holder of several patents. His main research interests include multispectral/hyperspectral image analysis, automatic object/target detection and recognition, pattern recognition, signal detection and estimation, and machine learning.

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